Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

Based on arxiv.org/abs/2504.04715

For ML researchers · AI security practitioners · Cloud infrastructure engineers

The Model Substitution Problem

Providers face strong economic incentives to covertly substitute cheaper alternatives — quantized versions, smaller models, or different architectures — to reduce GPU costs while maintaining pricing.

Formal Hypothesis Testing Framework

H_0: P_actual(y|x) = P_spec(y|x) vs H_1: P_actual(y|x) ≠ P_spec(y|x)

Smaller Model

e.g. 8B served

instead of 70B

Quantized Variant

FP8/INT8 instead

of FP16

Fine-tuned Version

Updated weights

without disclosure

Different Family

Entirely different

architecture

Black-box API access means users have no guarantee the advertised model is actually being served.

FP8 Quantization Preserves Most Accuracy

Performance drops are small — often within error bars — making substitution nearly undetectable via benchmarks alone.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Temperature Scaling Degrades All Models, Obscuring Substitution

At higher temperatures (0.5 – 2.0), performance drops across the board and the gap between original and FP8 variants narrows or becomes noisy, complicating statistical detection.

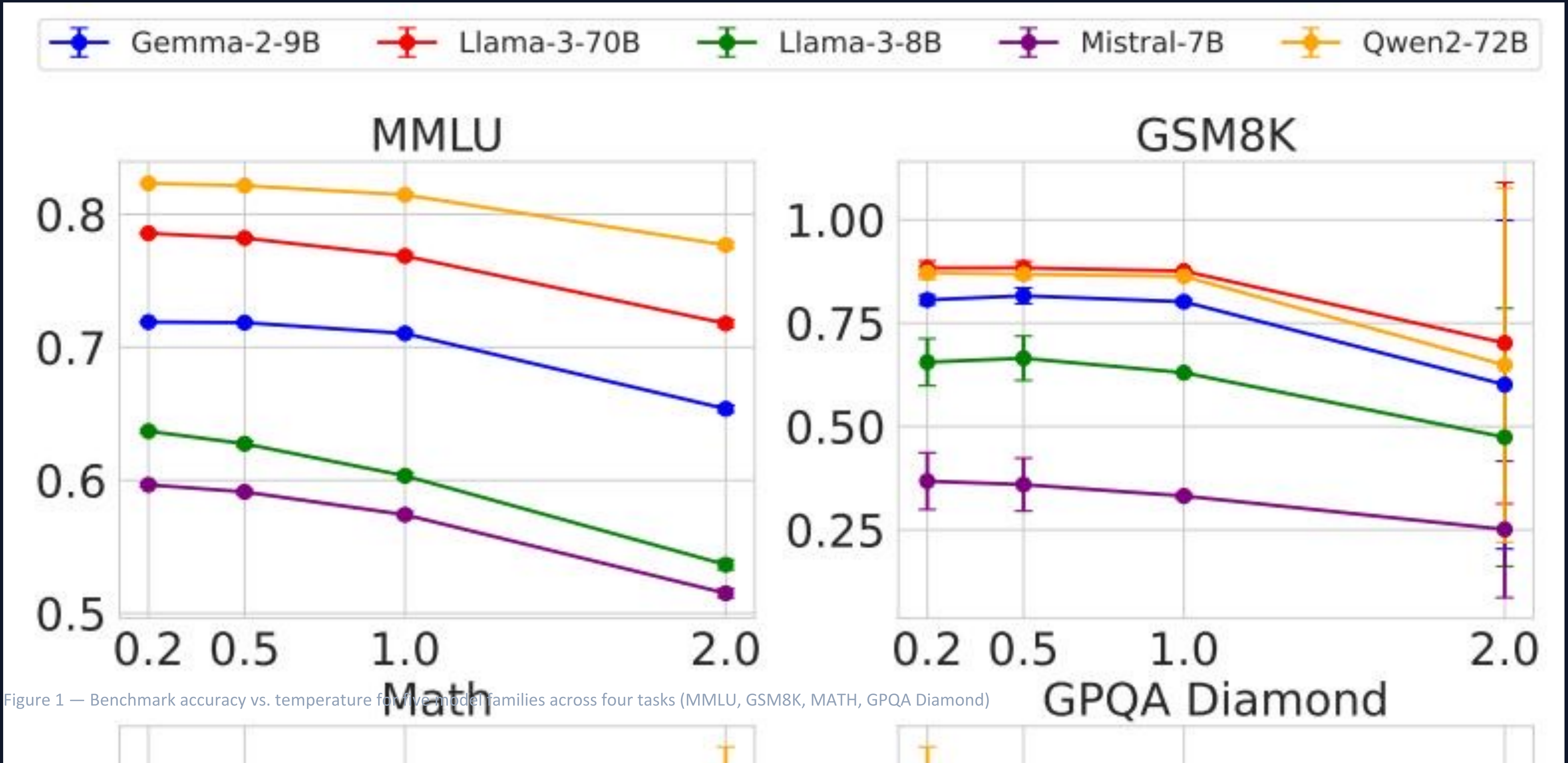

Figure 1 — Benchmark accuracy vs. temperature for five model families across four tasks (MMLU, GSM8K, MATH, GPQA Diamond)

Log-Probability Fingerprinting Fails Due to Inference Nondeterminism

The same Llama-3-8B model produces different log probabilities across software frameworks (transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2) and GPU hardware (A100 vs H100). Auditors cannot distinguish legitimate infrastructure variation from actual substitution.

Software Versions

transformers & vLLM version changes shift log-probs

GPU Hardware

A100 vs H100 produce measurably different outputs

Implication

Cannot separate infra noise from model substitution

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5 — Log-probability variation across inference frameworks and hardware for Llama-3-8B

Randomized Routing Makes Statistical Detection Query-Prohibitive

Providers can route only a fraction p of queries to a cheaper model. As p decreases, the mixed distribution converges to the specified model's distribution, requiring exponentially more queries for detection.

Mixed Distribution Model

$P\_mixed(y|x) = (1 - p) \cdot P\_spec(y|x) + p \cdot P\_sub(y|x)$ where $p \to 0$ makes detection infeasible

p = 50%

50% genuine / 50% substitute

Detectable with moderate queries

p = 10%

90% genuine / 10% substitute

Requires 25× more queries to detect

p = 1%

99% genuine / 1% substitute

Effectively undetectable

⚠ Benchmark Evasion: Providers can detect known benchmark prompts and route them to the genuine model, serving substitutes only for regular traffic.

Software-Only Auditing Methods Are Fundamentally Unreliable

Three independent failure modes demonstrate that no software-based approach can reliably detect model substitution.

Method Type	Core Weakness	Practical Limitation	Status
Text-Based Statistical Tests (e.g., MMD)	Requires impractically large query budgets	Fails against subtle or randomized substitutions	✗ UNRELIABLE
Log-Probability Fingerprinting	Defeated by production-level inference nondeterminism	Software versions & GPU hardware produce varying log-probs	✗ UNRELIABLE
Adversarial Counter-measures (combined)	Providers can actively evade any audit	Benchmark evasion + randomized routing make detection infeasible	✗ UNRELIABLE

Conclusion: Software-based detection is fundamentally limited — the auditor is always at an asymmetric disadvantage against a motivated provider.

TEEs Provide Provable Cryptographic Model Integrity

Trusted Execution Environments (TEEs) are hardware-secured enclaves that provide cryptographic attestation of the code and data being executed — proving that specific model weights are loaded and the correct inference code is running.

Provable

Cryptographic attestation guarantees model identity — not statistical inference, but mathematical proof.

Efficient

Modest performance overhead compared to unprotected inference. Practically deployable today via NVIDIA H100 confidential computing.

Robust

Immune to adversarial countermeasures. Cannot be evaded by randomized routing or benchmark detection.

Why not Zero-Knowledge Proofs?

ZKPs are computationally prohibitive for large-scale LLM inference. TEEs achieve the same trust guarantees with practical performance overhead.

NVIDIA Confidential Computing

H100 GPUs support hardware-level TEEs for inference, providing a practical deployment path available today for verifiable LLM API serving.

Key Takeaways: Close the Verification Gap with Hardware Attestation

1

Economic Incentive

Model substitution is economically motivated and practically undetectable via software — providers save significantly on GPU costs with quantized or smaller models.

2

Benchmark Blindness

FP8 quantization barely affects benchmark scores (often within error bars), making performance-based detection unreliable across MMLU, GSM8K, MATH, and GPQA Diamond.

3

Nondeterminism Kills Fingerprinting

Production inference nondeterminism across software versions and GPU hardware defeats log-probability fingerprinting — the same model looks different on different infrastructure.

4

Adversarial Evasion

Randomized substitution at low rates and benchmark-aware routing make all statistical detection methods query-prohibitive or entirely bypassable.

5

TEEs Are the Path Forward

Trusted Execution Environments provide provable, efficient, and robust model integrity guarantees — the only currently viable path to verifiable LLM APIs.